MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION NOVEMBER 2025

First Semester

Discipline Specific Course-MG1DSCCND100

BASIC NUTRITION AND DIETETICS

(2024 ADMISSION ONWARDS)

QP CODE: 25900776

Duration: 1 Hour Maximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability

Part A
Multiple ChoiceQuestions
Answer all questions
Each question carries 1 mark

1. 4 kcal

2. Carboxyl group

3. Food groups

4. Vitamin D

5. 65mg

6. Osteoclast

Part B
Short Answer Type Questions
Answer any 2 questions
Each question carries 2 marks

7. List out the enzymes involved in digestion of food in mouth.

- Salivary amylase (Ptyalin)- **1 Mark**
- Lingual lipase- **1 Mark**

8. **Summarize unsaturated fatty acids**

Unsaturated fatty acids - An unsaturated fat is a fat or fatty acid in which there is at least one double bond within the fatty acid chain. **(1 Mark)**

Example

Monounsaturated fatty acid (MUFA): A fatty acid chain is monounsaturated if it contains one double bond. Eg- Oleic acid **(0.5 mark)**

Polyunsaturated fatty acid (PUFA): A fatty acid is polyunsaturated if it contains more than one double bond. They are of 2 types- Omega-3 and omega-6 fatty acids. **(0.5 mark)**

9. Discuss on SCC

- SCC-Short Chain Carbohydrate
- They are soluble in 80% ethanol
- Eg:Raffinose,stachyose,poly dextrose,fructo-oligo saccharide (any 2)
- They are not hydrolyzed by human digestive enzymes.
- Fructo-oligo saccharides and Inulin stimulate the growth of bifido bacteria which is potentially beneficial to health.
  (Any 4 points -0.5 mark each)

10. Recall Diet counselling

- Diet counselling is a combination of nutrition expertise and psychological skills delivered by a trained dietitian. **(1 mark)**
- The process of diet counselling requires three basic skills namely interviewing, counselling and consulting. **(1 mark)**

Part C

Short Essay Type Questions

Answer any 4 questions

Each question carries 5 marks

11. Explain balanced diet.

- Balanced diet definition-**2 marks**
- Dietary sources of carbohydrates- Both digestible and non-digestible, protein- foods with high biological value, lipids- Vegetable source- rich in PUFA **-2 marks**
- Calorie distribution in a balanced diet **-1mark**

12. Explain the functions of carbohydrate.

- Source of energy
- Protein sparing action
- Oxidation of fats
- Indispensability for nervous system
- Role in muscle
- Role in liver
- Source of energy for heart muscle
- Synthesis of ribose
- Conversion of fat
- Promotes growth of desirable bacteria

- Contribution to dietary fibre
- Biosynthesis of amino acids
  (Any 5 points -1 mark each)

13. Discuss the functions and sources of Iodine

Functions

- Synthesis of thyroid hormones
- Development of brain
- Regulate the conversion of carotene into active vitamin A, the synthesis of protein and the absorption of carbohydrate from the intestine.
- Thyroxine is also known to be essential for reproduction. **(3 marks)**

Sources - Fish, seaweed, shrimp, and other seafood, which are generally rich in iodine. Dairy products and eggs, which are also good sources of iodine. Legumes and cereal grains are fair sources. **(2 marks)**

14. Discuss any five functions of phosphorus

- Mineralisation of Bones and Teeth
- Facilitation of Energy Transactions
- Absorption and Transport of Nutrients
- Regulation of Protein Activity
- Component of Essential Body Compounds
- Regulation of Acid-Base Balance **(Any 5 points -1 mark each)**

15. **Recall the RDA for folic acid different age groups**

Adult male	300 mcg
Adult female	220 mcg
Pregnancy	570 mcg
Lactation	330 mcg
Infants 0- 6 months	25 mcg
Infants 0- 6 months	85 mcg
Children 1-3 years	120 mcg
4-6 years	135 mcg
7 -9 years	170 mcg
Boys 10- 12 yrs	220 mcg
Girls 10-12 yrs	225 mcg
Boys 13- 15 yrs	280 mcg
Girls 13-15 yrs	245 mcg
Boys 16- 18 yrs	340 mcg
Girls 16-18 yrs	270 mcg

(RDA – 5 marks)

16. Discuss soft diet

- It is the second most common hospital diet. It bridges the gap between acute illness and convalescence. **(1 mark)**

- It may be used in acute infections, following surgery, and for patients who are unable to chew. The diet is made up of simple, easily digestible food and contains no harsh fibre, low in fat and no rich highly seasoned food. **(1 mark)**
- It is nutritionally adequate when planned on the basis of normal diet. Three meals with intermediate feedings are given in this diet. The diet provides 1500kcal and 35-40 g protein. **(1 mark)**

Mechanical soft diet is given for patients who have difficulty in chewing. It is limited to soft foods. **(0.5 marks)**

Fibre restricted soft diet differs from normal diet in being reduced in fibre content and soft in consistency. **(0.5 marks)**

Foods included are well cooked cereals, bread, ice-cream, milk, yoghurt, poached egg, mashed potato, fruits, pudding, soups etc. **(0.5 marks)**

Foods avoided are desserts like rich puddings and cakes, dried fruits and nuts, fried foods, highly seasoned soups, raw vegetables, spices and bran of cereals. **(0.5 marks)**

Part D

Essay Type Questions
Answer any 2 questions
Each question carries 10 marks

17. Review the functions of protein

- Building and tissue repairing / growth and maintenance of tissue
- Formation of essential body compounds- enzymes, hormones, haemoglobin, clotting factors, photoreceptors, tryptophan acts as a precursor of niacin
- Transporting nutrients
- Act as chemical messengers
- Regulation of water balance
- Maintenance of Appropriate pH
- Supporting the immune system
- Defence and Detoxification
- Providing energy
- DNA creation and Gene expression

(Each point carries 1 mark)

18. Discuss the roles of BMR, SDA, PAL and TEE in nutritional planning and weight Management strategies.

- **BMR-** Definition – **1 Mark**
- Forms the **foundation** of energy requirement calculations - **0.5 Marks**

- Helps determine the **minimum calorie intake** needed to sustain life- **0.5 Marks**
- Individuals with higher BMR require more calories; those with lower BMR need fewer- **0.5Marks**
- Knowing BMR helps create calorie deficits for weight loss or surpluses for weight gain safely- **0.5 Marks**

- **SDA-** Definition – **1 Mark**
- Contributes to overall calorie burn; protein-rich foods have a higher SDA, which can slightly increase energy expenditure- **0.5 Marks**
- Important in planning balanced diet for weight management. - **0.5 Marks**

- Aids in understanding how meal frequency and food choices influence metabolism. - **0.5 Marks**

- **PAL** - Definition – **1 Mark**
- The **most variable** part of energy expenditure; crucial for customizing diet plans. - **0.5 Marks**
- Higher PAL increases energy needs, essential for athletes or active individuals. - **0.5 Marks**
- Enables dietitians to tailor activity recommendations—walking, exercise routines, or lifestyle modifications. - **0.5 Marks**

- **TEE-** Definition – **1 Mark**

- Helps to determine whether a person should consume: Less than TEE → for weight loss, Equal to TEE → for weight maintenance, More than TEE → for weight gain- **0.5 Marks**

- Essential for calculating precise caloric targets based on age, sex, body composition, health status, and lifestyle- **0.5 Marks**

19. **Discuss the functions, sources and requirements of Niacin**

Functions

- Release of energy from CHO, protein, fat and alcohol
- Synthesis of protein, fat and pentose sugars for formation of RNA and DNA
- Part of various coenzymes, the coenzymes help in the biological oxidation and in metabolic reaction of CHO.

(3 marks)

Sources - Whole cereals, pulses, nuts and meat are good sources. Groundnut is particularly rich in niacin. **(2 marks)**

Requirements **(5 marks)**

Adult male sedentary	14 mg

	18 mg
	23 mg
Adult female	11 mg
	14 mg
	18 mg
Pregnancy	+ 2.5 mg
Lactation	+0.5 mg
Infants 0- 6 months	2 mg
Infants 0- 6 months	5 mg
Children 1-3 years	7 mg
4-6 years	9mg
7 -9 years	11 mg
Boys 10- 12 yrs	15 mg
Girls 10-12 yrs	14 mg
Boys 13- 15 yrs	19 mg
Girls 13-15 yrs	19 mg
Boys 16- 18 yrs	22 mg
Girls 16-18 yrs	17 mg

20. Explain the types of Dietitian

Dietitians are the health professionals, ideally trained to implement and change dietary habits of an individual or population. **(Any definition – 1mark)**

- Clinical dietitian
- Consultant dietitian
- Community dietitian
- Research dietitian
- Neonatal dietitian
- Pediatric dietitian
- Gerontological dietitian,
- Business dietitian
- Administrative dietitian **(Each Classification carries 1 mark)**